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The spatio-temporal patterns of landfast ice in Antarctica during 2006–2011 and 2016–2017 using high-resolution SAR imagery

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ABSTRACT

Landfast ice is an important component of the Antarctic sea ice regime. It affects the Antarctic climate and ecological system. In this study, the first high-resolution, long time series of the landfast ice edge from 2006 to 2011 and 2016 to 2017 is presented. The dataset was produced based on the improved net gradient difference algorithm using 2470 SAR scenes from ENVISAT and Sentinel-1A/B as well as manual analysis of MODIS imagery to fill in SAR data gaps. The study results show that the landfast ice area in November for all studied years was approximately $49.49 \pm 3.25 \times 10^4$ km^2, accounting for about 3%–4% of the total Antarctic sea ice area. The maximum area was 55.70×10^4 km^2 in November 2007, compared to the minimum area 44.01×10^4 km^2 in 2011. The area in West Antarctica was about 40% of that in East Antarctica. The distribution of landfast ice in Antarctica has significant regional differences. The extent in the Indian Ocean sector is the maximum with a mean value of $16.49 \pm 1.1 \times 10^4$ km^2; however, the ratio of the landfast ice area to the sea ice area in the Pacific Ocean sector is the highest. Twenty-four landfast ice zones with groups of small, grounded icebergs were identified, most of which were located in East Antarctica, particularly along the Wilkes Land and Oates Land. Two cases are presented to illustrate how giant, grounded icebergs affected landfast ice. Results from this study are well suited to underpin the Antarctic climate or ecological system studies.

1. Introduction

Fast ice is the sea ice that remains stationary by being attached to the shoreline, glacier tongues, ice shelves, ice walls, grounded icebergs, or grounded over shoals (Cheng Bin et al., 2008; Massom et al., 2001). It is commonly found in many parts around the nearly 35,000 km coast of Antarctica (Yu et al., 2019), which features numerous bays, fjords, ice shelves and grounded icebergs. Fast ice may exist only during the freezing season, with its largest extent observed by the end of the season, or it may linger for several years. It extends from tens to hundreds of kilometers from its anchor point, with a thickness that may reach 2 m (Price et al., 2014; Zhao et al., 2017). Part of this thickness is likely formed from frazil ice crystals (platelet ice) collected under the ice shelves (in the supercooled water resulting from the ice shelf basal ablation) when favorable ocean current transports and deposits them under the existing fast ice (Langhorne et al., 2015). Fast ice breaks under the oceanic force (high wave) or wind force, the latter (katabatic wind) is more common in the coastal areas of the Antarctic. When it breaks under wind action, it may generate coastal polynya, a phenomenon commonly observed at the edge of fast ice. The fast ice extent depends on a few factors. These include the wind (may support anchoring or breaking of the ice), air temperature (below-freezing values accelerate the ice formation), water circulation, and tide. Combinations of these factors lead to interannual variability in the landfast ice extent.

As a recurrent phenomenon, landfast ice plays an important role in the cryosphere environment (Hunke et al., 2010). Its extent, timing and thickness affect the heat flux and moisture exchange between the air and ocean (Heil, 1996; Mahoney et al., 2007; Murphy et al., 1995),

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thereby modulating the Antarctic climate. Landfast ice plays a role in the formation and maintenance of polynyas around Antarctica (Fraser et al., 2010; Massom et al., 2001; Nihashi and Ohshima, 2015), where high ice production occurs. With it, the potential for the Antarctic Bottom Water is enhanced. This is a major driving force for the global overturning circulation (Marshall and Speer, 2012; Takeshi Tamura et al., 2008; Young et al., 1998). Recent studies have shown that landfast ice can enhance the stability of ice shelves/tongues by protecting them from the ocean's swell and waves, and therefore affects the mass balance of the Antarctic ice sheet (Massom et al., 2010; Miles et al., 2017). In addition, landfast ice provides a crucial habitat and hunting platform for aquatic animals (e.g. emperor penguin and seals) and marine organisms (Linial, 2015; Massom et al., 2009; Trathan et al., 2011).

Because of these significant roles of landfast ice in the Antarctic climate and ecological systems, a comprehensive understanding of its spatio-temporal variability is necessary. However, the harsh environment of the Antarctic, particularly during the polar night, has restricted field measurements of landfast ice. In situ observations have been carried out nearby some research stations at various locations, such as Davis Station (Heil, 1996, 2006), Lutzow-Holm Bay where Syowa Station is located (Enomoto et al., 2002; Ushio, 2006), McMurdo Sound near McMurdo Station (Kim et al., 2018; Wongpan et al., 2018), and Zhongshan Station (Lei et al., 2010). However, these field measurements are too sparse considering the length of the Antarctic coastline.

Remote sensing provides an efficient way to study landfast ice in the Antarctica region. Observations by ENVISAT ASAR (Advanced Synthetic Aperture Radar) images acquired between November 2002 and March 2012 indicated that two glacier calving events were driven by the break-up of the landfast ice, which usually occupied the Porpoise Bay, Wilkes Land (Miles et al., 2017). Advanced Very High Resolution Radiometer (AVHRR) and Moderate Resolution Imaging Spectrometer (MODIS) data have been employed to study the landfast ice distribution and monitor break-up events in the Lutzow-Holm Bay (Aoki, 2017; Ushio, 2006). The combination of MODIS data and sea ice concentration data derived from the Scanning Multichannel Microwave Radiometer (SMMR) and the Special Sensor Microwave Imager (SSMI) were used to study the local climatology of landfast ice in McMurdo Sound, combined with ice thickness, temperature and wind velocity (Kim et al., 2018). A time series of the landfast ice extent off the Ade'lie Land coast over 134°–143°E from 1992 to 1998 was created from cloud-free AVHRR data to investigate the interannual variability of landfast ice and its implications for emperor penguins (Massom et al., 2009).

At present, there are only a few large-scale or hemisphere-scale studies on landfast ice in Antarctica. Two “snapshot” maps of the landfast ice extent in East Antarctica, which covered 87.8°–170.3°E in November 1997 and 75.15°–150.16°E in November 1999, were obtained using Radarsat-1 SAR image by image cross-correlation technique (Giles et al., 2008). The first time-series dataset of the landfast ice extent in East Antarctica during 2000–2008 was created using consecutive cloud-free 20-day MODIS composite images with 1-km resolution (Fraser et al., 2010; Fraser et al., 2012). The circumpolar mapping of landfast ice and polynya of Antarctica were generated using the Advanced Microwave Scanning Radiometer for Earth Observing System (EOS) (AMSR-E) data with a 6.25-km resolution (Nihashi and Ohshima, 2015), which revealed the significant relationship between landfast ice and coastal polynyas. All these studies have advanced the Antarctic landfast ice mapping. However, long time series and high-resolution landfast ice dataset are still lacking.

In this study, we employed an automatic method (Li et al., 2018) to detect the landfast ice edge (LIE) at the high resolution of SAR around Antarctica. It was developed based on the net gradient difference algorithm presented by Mahoney et al. (2004) and Mahoney et al. (2007). The ENVISAT ASAR and Sentinel-1A/1B data acquired during 2006–2011 and 2016–2017 were used to observe the maximum landfast ice extent (usually observed in November) along the entire Antarctic coast.

2. Data

Remotely sensed data and the Antarctic coastline data are used in this study. The remotely sensed data include ENVISAT ASAR images and Sentinel-1A/1B SAR images for detecting the LIE as well as cloud-free MODIS true-color images for validation and filling gaps in SAR data. The Antarctic coastline data include the coastline derived from the MODIS Mosaic of Antarctica (MOA) in 2009 (Scambos et al., 2007) and a new dataset developed from the combination of Landsat-7/8 data and ASAR data for2005, 2010 and 2015 (Yu et al., 2019). Sea ice concentrations were derived from AMSR-E/2 data and the Bedmap-2 bathymetric data and icebergs tracking data were used as auxiliary datasets to support the analysis of variation in landfast ice.

2.1. SAR dataset

Two SAR data sets from the European Space Agency (ESA) were used: ENVISAT ASAR data acquired during 2006–2011 and Sentile-1A/1B SAR data acquired during 2016–2017. Both datasets were acquired in November when the landfast was at its maximum extent (end of the austral spring). SAR image acquisition dates and characteristics are presented in Table 1. ASAR operated in the C-band (central frequency: 5.331 GHz) with a 35-day repeat orbit and 5 distinct measurement modes with different spatial resolutions and coverages (ESA, 2007). Data for this study were obtained in the Wide Swath (WS) mode (approximately 405 km) at a medium resolution (150 m) utilizing the ScanSAR technology. Sentinel-1A/B is the satellite constellation of ESA's Copernicus program, with a C-band SAR (central frequency: 5.405 GHz) that features a 12-day repeat cycle for each satellite and four modes of radar operation Stripmap (SM), Interferometric Wide swath (IW), Extra-Wide swath (EW), and Wave (WV) (ESA, 2016). We used S1A/1B's EW Ground Range Detected Medium (GRDM) products, which are based on the TopSAR technology (Zan and Guarnieri, 2006), owing to its high spatial resolution (40 m) and large swath width (400 km).

Based on the automatic net gradient difference algorithm for edge

Table 1
SAR images used in this study.

Sensor	Year	Period 1		Period 2		Period 3		Incidence angle	Pixel spacing (m)
		Acquisition date	No. of images	Acquisition date	No. of images	Acquisition date	No. of images		
ASAR WSM	2006	Nov 8–Nov14	73	Nov 15–Nov 21	109	Nov 22–Nov 30	100	15°–45°	75 × 75
	2007	Oct 18–Oct 24	63	Oct 25–Oct 31	102	Nov 1–Nov 7	80		
	2008	Nov 8–Nov14	54	Nov 15–Nov 22	134	Nov 23–Nov 30	116		
	2009	Nov 8–Nov14	112	Nov 15–Nov 21	109	Nov 22–Nov 28	137		
	2010	Nov 8–Nov14	73	Nov 15–Nov 21	60	Nov 22–Nov 28	57		
	2011	Nov 8–Nov14	95	Nov 15–Nov 21	92	Nov 22–Nov 28	102		
Sentinel-1 EW	2016	Nov 8–Nov14	130	Nov 15–Nov 21	127	Nov 22–Nov 28	136	18.9°–47°	40 × 40
	2017	Nov 8–Nov14	137	Nov 15–Nov 21	137	Nov 22–Nov 28	135		

detection (Li et al., 2018), three consecutive images obtained within approximately 20 days are needed. Therefore, individual images acquired within 7 days should be employed to compile a mosaic covering the Antarctic coast as completely as possible. In the Antarctic region, the landfast ice starts to melt by the end of November after it reaches its maximum extent (Lei et al., 2010). The observations used in this study were obtained to generate mosaics for three sequential time periods, period 1 for approximately November 8 to November 14, period 2 for November 15 to November 21 and period 3 for November 22 to November 28. Images used to create a mosaic should fulfill three criteria: (1) the output mosaic should cover the Antarctic coast as completely as possible; (2) there should be minimum overlap among two adjacent images, and (3) repeat-pass images during any given period are removed to reduce the data processing cost. As a result, we have selected 1668 ASAR WSM images for 2006–2011 and 802 Sentinel-1A/1B EW images for 2016–2017 to assemble the three-period Antarctic mosaic images (Table 1). In 2007, many ASAR images were acquired in the middle-late October and early November owing to the lack of sufficient images in the middle-late November.

The orbits of the selected images over the Antarctic coast in 2011 are plotted in Fig. 1. Data gaps can be seen along Victoria Land and Wilkes Land in periods 1 and 3. Gaps also exist in other years.

2.2. MODIS images

To get the full coverage of the Antarctic coast, MODIS cloud-free true-color images (250 m resolution from visible bands) were obtained from the National Aeronautics and Space Administration's (NASA) Worldview and used to fill the gaps in SAR data (gaps are shown in Fig. 2 for each year). MODIS visible imagery obtained in November in various years, was used to generate LIE manually in areas where gaps in SAR data were encountered. A MODIS visible image comprises three bands: 1 (645 nm wavelength), 4 (555 nm wavelength) and 3 (469 nm wavelength), which are combined in the RGB composite color, respectively after applying the image sharpening technique (Zhao et al., 2017).

2.3. The Antarctic coastline data

In Antarctica, landfast ice is attached to the coast or to permanent ice entities such as an iceberg or ice shelf. Accurate coastline datasets for various years are required to define the inner boundary of the landfast ice and its area. Along the Antarctic coast, the advancing or calving of ice shelves and ice tongues leads to changes in the coastline (Ferrigno et al., 2008; Ferrigno et al., 2005). In this study, the Antarctic coastline data was used to represent the inner (near-shore) boundary of landfast ice as follows. The coastline datasets were not available for each of the study years in this work. Therefore, coastline datasets sufficiently close to each of the study years were used. For landfast of 2006 and 2007, the coastline of 2005 was used. For 2008 and 2009, the coastline of 2009 was use. For 2010 and 2011, the coastline of 2010 was used. For 2016 and 2017, the coastline of 2017 was used. Coastline data of 2005, 2010 and 2017 were derived from Landsat-7/8 and ASAR data using an automated edge detection method (Yu et al., 2019). The location precision of these three coastlines is estimated to be no worse than ±30 m, better than one pixel of the SAR image used in this study. On the other hand, the coastline of 2009 was manually digitized from the 125-m MODIS MOA image and provided by the National Snow and Ice Data Center (NSIDC) (Scambos et al., 2007). The location precision of this coastline is estimated to be no worse than ±250 m.

2.4. Other auxiliary data

Daily sea ice concentration data were generated at the University of Bremen using ASI (The Arctic Radiation and Turbulence Interaction Study (ARTIST) Sea Ice) algorithm (Spreen et al., 2008). Input data to the algorithm constitute observations from the passive microwave radiometer Advanced Microwave Scanning Radiometer-EOS/2 (AMSR-E/2). Sea ice concentration data was used in this study to analyze the relation between the total sea ice and landfast ice areas. First, the monthly mean sea ice concentration was calculated using the daily sea ice concentration data for November. The total sea ice area, defined as the sum of the products of the ice concentration and pixel area for all pixels with concentration larger than 15%, was then calculated. The Bedmap-2 data (Fretwell et al., 2013) provided by the British Antarctic Survey was used for the spatial analysis of landfast ice.

3. Method

3.1. SAR mosaic image

SAR data was preprocessed to obtain a mosaic image of the Antarctic contour (including the landfast ice coverage). This included geometric correction, radiometric calibration and geocoding projection. During geometric correction, the precise-orbit files provided by ESA were used to achieve an accurate geolocation within one pixel accuracy or better (Schubert et al., 2017; Schubert et al., 2008). Radiometric calibration was then performed to convert the digital number (DN) values into a backscatter coefficient (in decibel). Subsequently, all images were geocoded with the South Polar Stereographic projection. Next, the pixel size of the Sentinel-1A/B image was resampled to 80 m to ensure a similar spatial resolution to ASAR WSM data. These preprocessing steps were performed using the SARscape software (in ENVI) for ASAR WSM images and Sentinel Application Platform (SNAP) software for the Sentinel-1A/B EW-GRD image.

In Antarctica, the distance of LIE to the coastline is generally <150 km (Giles et al., 2008). Therefore, a buffer with a 250-km distance off the coastline was adopted to clip the mosaic image of fast ice

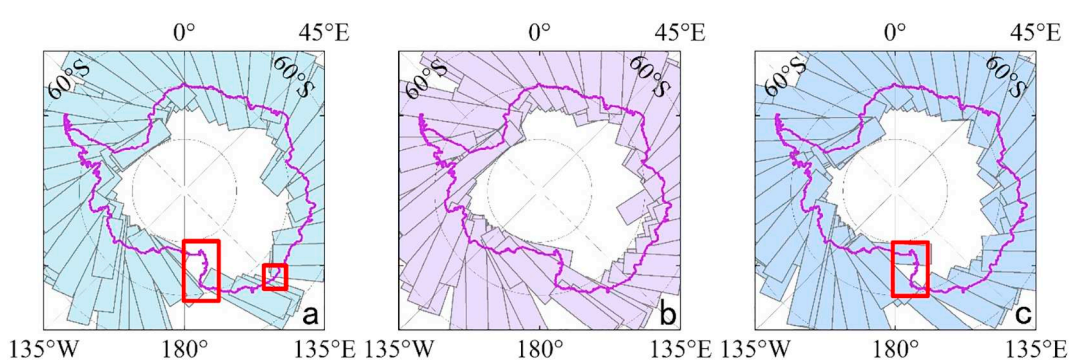

Fig. 1. Coverage orbits of ASAR images acquired in November 2011. Panels a, b, and c, correspond to periods 1, 2, and 3, respectively (Table 1). Rectangles represent spatial coverages of individual images. Red frames in a and c denote the gaps in ASAR coverage. Purple curve is the Antarctic coastline in 2010 (Yu et al., 2019). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

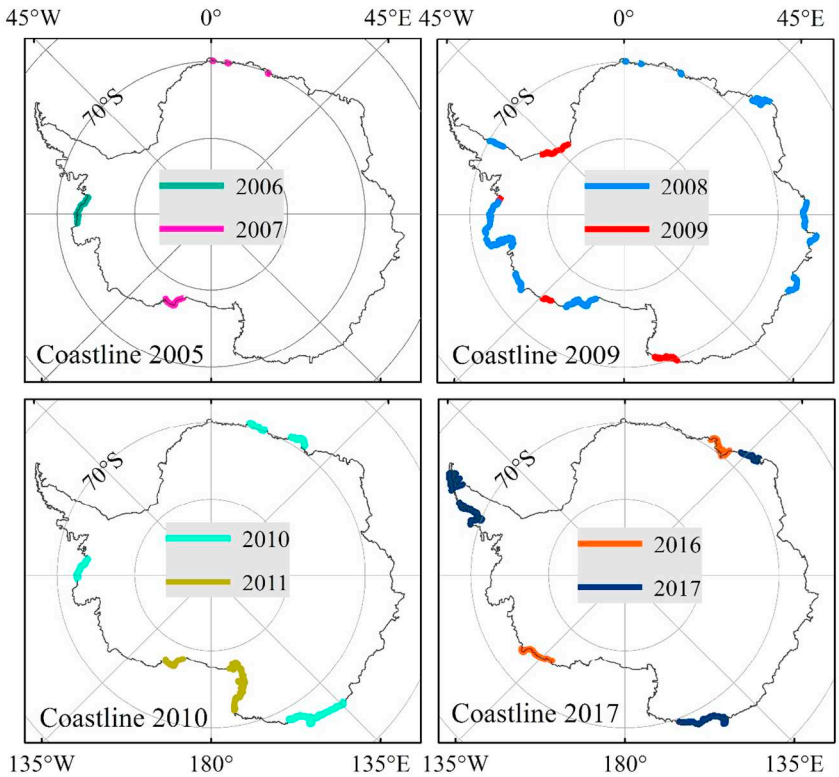

Fig. 2. Distributions of all gaps in SAR data during the study period.

around Antarctica. This had to be performed to reduce the time and cost of computation.

3.2. Landfast ice edge detection

The LIE was detected using the improved net gradient difference algorithm (Li et al., 2018), which is based on the gradient difference algorithm (Mahoney et al., 2004). It employs spatial and temporal gradient operators to identify potential LIE pixels, a method for removing noise pixels, and a method to connect the edge pixels. This approach suits the LIE definition at the higher spatial resolution of SAR in winter when the surface backscatter from landfast ice is relatively stable.

The algorithm comprises the following steps. First, the two-dimensional spatial gradient vector (i.e. horizontal and vertical components) is calculated for each SAR image. Subsequently, the temporal gradient of the vertical and horizontal spatial gradient components is computed using each component of spatial gradient vectors calculated from three consecutive images (mosaics) within the periods shown in Table 1 for each year. Next, the net gradient difference is calculated using the vertical and horizontal temporal gradients. After that, the raster image of the LIE is generated by applying the medium edge detection method (Zhou et al., 2010). Finally, the LIE is generated by vectorizing its points and removing noise (Li et al., 2018).

During the implementations of the algorithm, upper and lower threshold (i.e., the minimum and maximum locations of the LIE obtained from data for previous years) were used to constrain the derived LIE to be within that range. The range was determined as the average of the maximum and minimum landfast ice extents derived from MODIS data for day of year (DOY) 281–300 in years 2000–2008 in East Antarctica (10°W–172°E) (Fraser et al., 2012). No such reference data is available for West Antarctic. Therefore, the range was manually delineated in this region. For gaps in the SAR coverage, LIE was obtained using visual interpretation and manual depiction from cloud-free true-color MODIS images (250 m resolution). No incidence angle correction of SAR images was applied in our study.

3.3. Validation

The validation of the output LIE was performed in a previous study (Li et al., 2018) by comparison against manual extraction of the edge using SAR data as well as a near-coincident Landsat-8 OLI image. In the present study, further validation is offered by overlaying the SAR-derived LIE output on the MODIS images where LIE was visually depicted and consequently used as a reference. Four sections were used as examples: the Lutzow-Holm Bay (LHB), West Ice Shelf (WIS), and Mertz Glacier (MG) in 2009 and Filchner Ice Shelf (FIS) in 2017 (Fig. 3). Qualitatively, the two LIE sets (from SAR and MODIS) are in close proximity to each other (Fig. 3). The cloud-free MODIS images were acquired in late November or early December, which means that there could be a lag of a few days in the acquisition date between MODIS and ASAR/Sentinel-1A/B.

A quantitative evaluation of the SAR-generated LEI was performed by comparing the locations of a set of points with a 0.1° longitude interval from the SAR-driven LIE against the corresponding points from the subjectively delineated LIE in MODIS. The distance (deviation) between each longitude pair was calculated. Table 2 shows the average deviation for the aforementioned four areas (recall the coarser resolution of MODIS data). The deviation is <0.7 km for the LHB, MG and FIS sections and slightly above 1 km for the WIS section. The maximum deviation of 24.73 km was observed in the WIS within the red frame marked in Fig. 3b. At this location, the three dates used to derive the temporal gradient from ASAR data were November 12, 19 and 22, 2009. This is about 3 weeks prior to the acquisition of MODIS on December 3, 2009. This significant deviation (24.73 km) is likely caused by the breakoff of fast ice during this three-week period. The exact day could not be identified in MODIS data owing to the heavy cloud over this area during this period. If the pair of points within the red frame is excluded from the statistics, the average deviation for the WIS area drops to 0.09 km (from 1.01 km). Data from this area after excluding

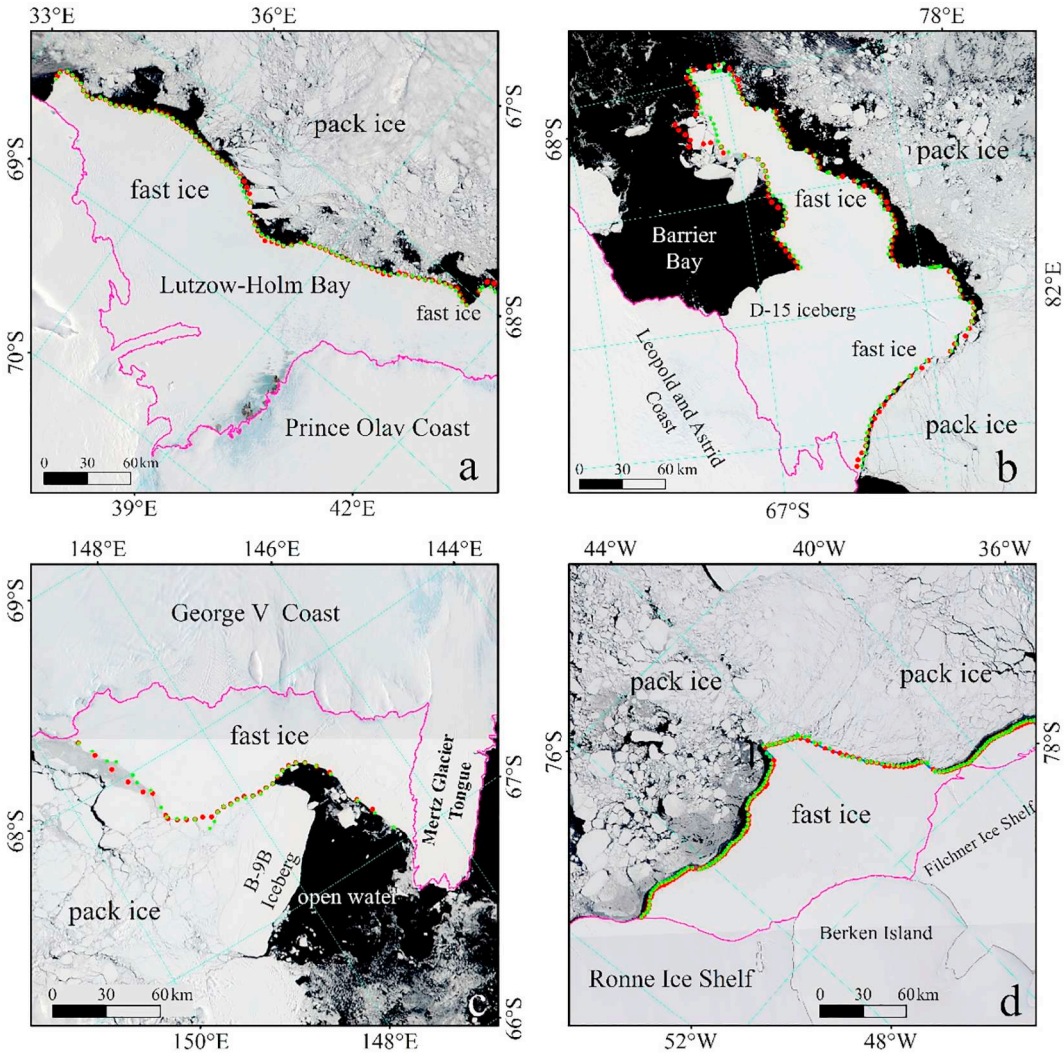

Fig. 3. LIE detected from SAR images overlaid on visible MODIS images (bands 1, 4, 3) of (a) LHB (November 27, 2009), (b) WIS (December 3, 2009), (c) MG (November 29, 2009) and (d) FIS (November 22, 2017). The red points and green points are samples of the LIE algorithm output from SAR and manual edge identification from MODIS images, respectively. Purple line is the Antarctic coastline (2009 or 2017). Red frame in (b), where large discrepancy is observed is a region where fast ice broke off earlier (exact day was not determined because of unavailability of cloud-free MODIS data). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 2
Difference (in km) between automated LIE obtained from ASAR data and the manual analysis of MODIS data for the areas and dates shown in Fig. 3. Positive values indicate that LIE from SAR is farther away from the shoreline.

	LHB	WIS	MG	FIS	WIS*	Mean1[a]	Mean2
Average	0.27	−1.01	−0.39	−0.71	0.09	−0.46	−0.18
Min	−2.86	−24.73	−7.42	−5.21	−6.03	−10.05	−5.38
Max	11.18	6.33	4.83	2.56	6.33	6.23	6.23

[a] Mean1 is the mean value off LHB, WIS, MG and FIS. Mean2 is the mean value of LHB, WIS*, MG, and FIS.

these anomaly points are presented in the column WIS* in Table 2. The total mean deviation also decreased from −0.46 km to −0.18 km.

The landfast ice area of the four sections was also calculated from the same SAR and MODIS data sets (Table 3). The percentage of area difference is minimum for LHB (0.3%) and maximum for WIS (4.68%) (because of the fast ice breakup event). The average percentage after excluding the breakup event is 0.66%. We consider this a reasonable difference that establishes acceptable accuracy of the automated LIE method used in this study.

Table 3
Comparison of landfast ice areas (km^2) between SAR and MODIS.

	LHB	WIS	MG	FIS	WIS*	Mean1[a]	Mean2
SAR	45,375	19,495	9849	18,068	19,015	23,197	23,077
MODIS	45,238	18,624	9695	18,315	18,624	22,968	22,968
Difference	137	871	154	−247	391	229	109
Percentage	0.30	4.68	1.59	−1.35	2.10	1.30	0.66

[a] Mean1 is the mean value of LHB, WIS, MG and FIS. Mean2 is the mean value of LHB, WIS*, MG and FIS. Percentage is calculated as difference/MODIS ∗ 100%.

4. Results

4.1. Spatial distribution of landfast ice

Mapping of landfast ice during the study period is presented for five sectors of the Southern Ocean: Weddell Sea (WS, 20°E–55°W), Bellingshausen and Amundsen seas (BAS, 55°–130°W), Ross Sea (RS, 130°W–153°E), Pacific Ocean (PO, 153°E–100°E) and Indian Ocean (IO, 100°E–20°E). These sectors are affected by different geographies and

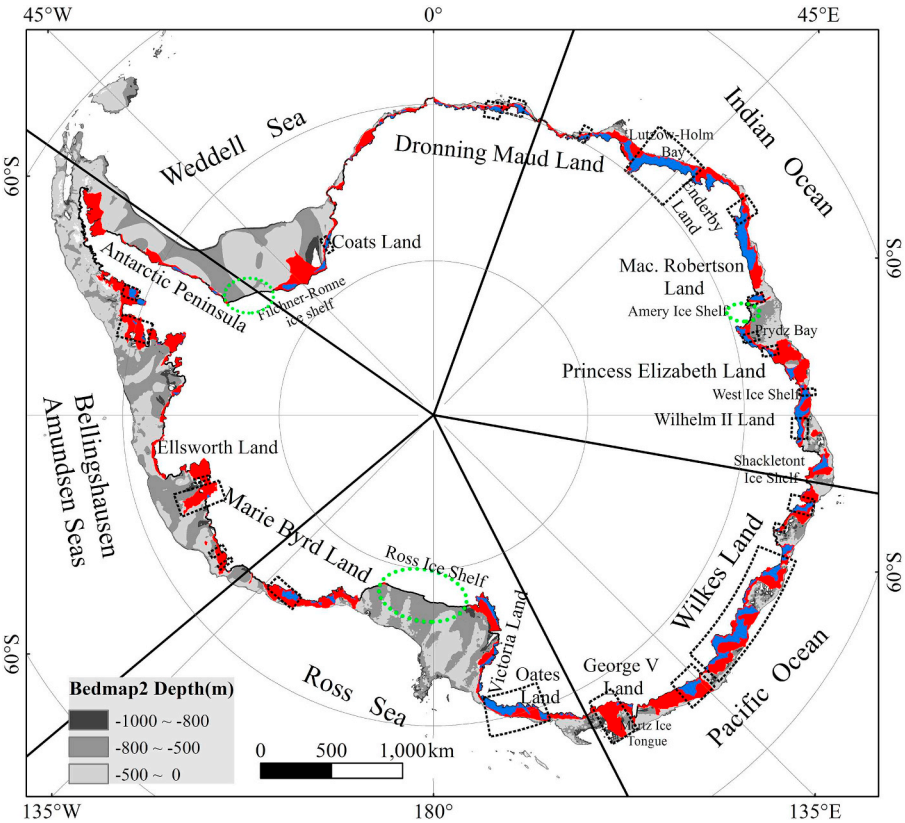

Fig. 4. The circum-Antarctic distribution of landfast ice. Red regions represent areas where landfast ice existed at least once in November during the study period. Blue were recurrent landfast ice which existed every year in November during the study years. Bathymetric data from Bedmap2 is displayed. Green dashed ellipses mark the front of Ronne Ice Shelf, Ross Ice Shelf and Amery Ice Shelf. Black dashed boxes represent 24 landfast ice zones with groups of grounded small icebergs identified in SAR images. Black lines are boundaries of the five sectors. The coastline and land mask in the figure are from 2005 data based on Yu et al. (2019). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

weather conditions and have been used in previous Antarctic sea ice studies (Cavalieri and Parkinson, 2008; Parkinson and Cavalieri, 2012). The sectors and the delineated LIE for the eight studied years in each sector are presented in Fig. 4.

During the study period, landfast ice existed along most of the Antarctic coast in November (red regions in Fig. 4). Nevertheless, there was no landfast ice in front of Ronne Ice Shelf, Ross Ice Shelf and Amery Ice Shelf (green ellipses in Fig. 4), which are the three largest ice shelves in the Antarctic. Polynyas are dominant in front of these ice shelves because of the strong katabatic wind (Nihashi and Ohshima, 2015). While sea ice continues to form in front of these shelves, the strong katabatic wind blows it away and prevents it from stabilizing and becoming fast. In general, landfast ice tends to reoccur at specific locations every year. The blue regions in Fig. 4 show the distribution of the recurrent landfast ice around entire Antarctica; the landfast ice appeared in these areas every year during the study period. It can be seen that the Indian Ocean and Pacific Ocean sectors (30°–140°E) retain the most landfast ice. In addition, there is also significant landfast ice along the Victoria Land and Oates Land, the west coast of Ross Sea. Less recurrence of landfast ice is observed off West Antarctica, with less extent than off East Antarctica, which indicates the strong inter-annual variation in West Antarctica than in East Antarctica.

As shown in Fig. 4, landfast ice generally occurs where the bathymetry is <500 m. The average depth of the Antarctic continental shelf is about 500 m, which is usually the maximum depth where icebergs can be grounded (Beaman and Harris, 2005; Li et al., 2017). The width of the continental shelf determines the northern border of landfast ice in East Antarctica. The continental shelf is wider in West Antarctica than in East Antarctica. However, LIE does not reach the edge of the continental shelf. It appears that the factors affecting the position of the northern border of landfast ice might be different in East than in West Antarctica. In the future, more observations and modeling data should be obtained to analyze the relationships between the LIE and bathymetric contour.

Fig. 5 shows the shortest distance from the LIE to the coastline of Antarctica for each year. In this study, the shortest distance is used to represent the width of the landfast ice. For entire Antarctica, the width of landfast ice ranged from a few kilometers to >200 km. Most of the widths are <150 km, which is in agreement with previous studies (Giles et al., 2008; Massom et al., 2001). The Indian Ocean and Pacific Ocean sectors have landfast ice zones with width between 50 km and 100 km. The mean widths of landfast ice during the study years in these two sectors were 49.2 km (IO) and 56.4 km (PO), respectively. In the WS sector, the width of the most of landfast ice was <30 km, and its mean width was 25.6 km. However, there was a landfast ice zone with width approximately 200 km from 2006 to 2009, which decreased to below 100 km beyond 2010. This is analyzed in case study 2 (Section 5). The mean widths of landfast ice in the BAS sectors was also 25.6 km and the mean width in the RS sector was 29.1 km.

4.2. Interannual variation in landfast ice

The total sea ice area, defined as the sum of the products of the ice concentration and pixel area for each pixel, was calculated using AMSR-E/2 sea ice concentration data. Table 4 shows the landfast ice area and its ratio to the total sea ice area during 2006–2011 and 2016–2017 in the five sectors and the East and West Antarctic regions (EA and WA, respectively). During the study years, the mean landfast ice area was 49.49×10^4 km^2. The minimum area was 46.66×10^4 km^2 in 2010 and the maximum area was 55.70×10^4 km^2 in 2007. The landfast ice areas in the first four years were larger than those in the last four years. The IO sector had the largest landfast ice area, with a mean of $16.49 \pm 1.1 \times 10^4$ km^2, followed by the PO sector with a mean of $13.23 \pm 1.96 \times 10^4$ km^2. Both sectors are located in East Antarctica. The BAS and RS have similar landfast ice areas, approximately half of that in IO sector. The WS sector has the smallest landfast ice area, which is approximately a quarter of that in the IO sector. The West Antarctic region has less landfast ice area than the East Antarctic (ratio is <0.5)

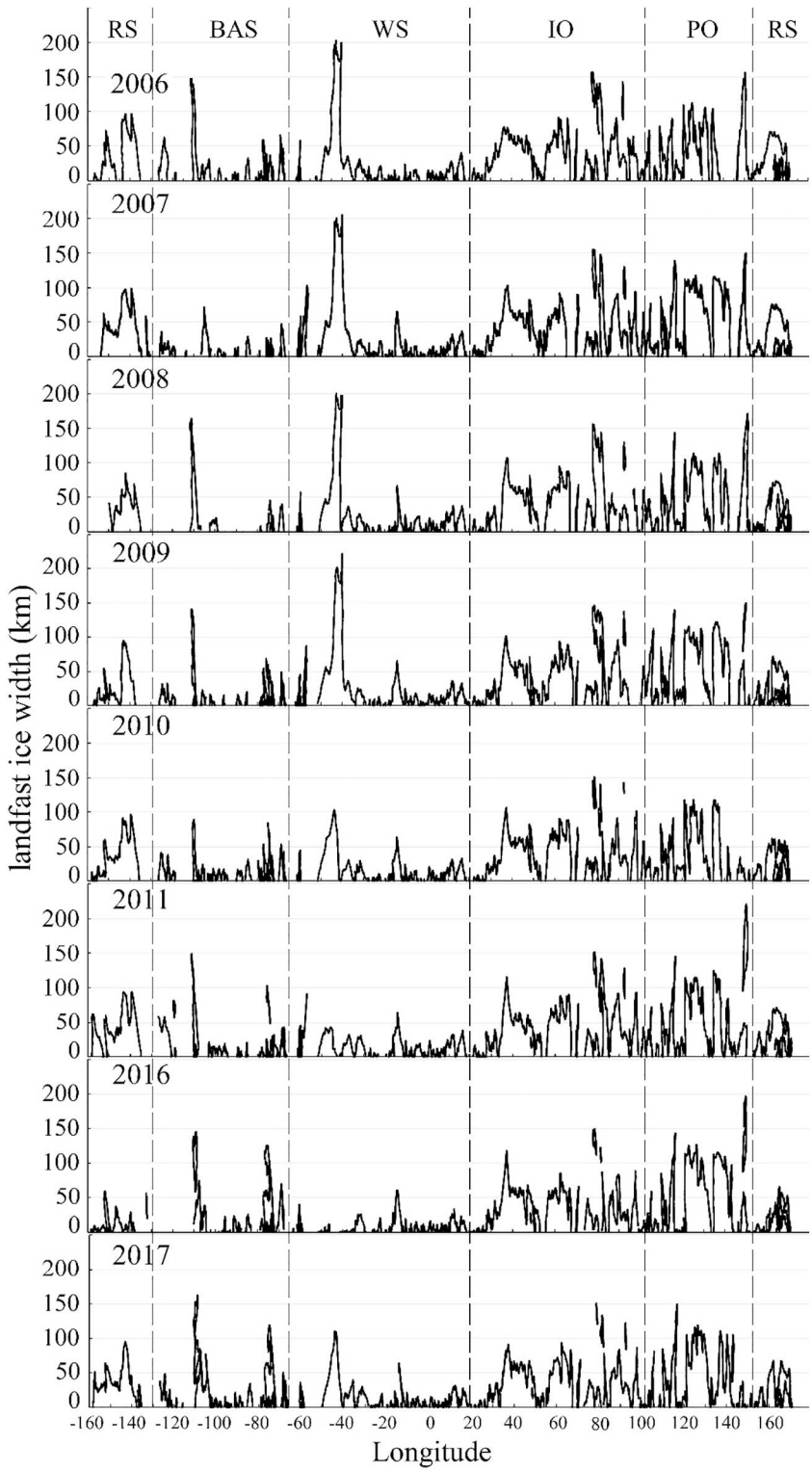

Fig. 5. Landfast ice width distribution along the Antarctic coast for each year. The x-axis represents the longitude. Black dashed vertical lines are boundaries of five sectors.

with remarkably smaller ratio (0.38) in years 2008 and 2011.

The pack ice and fast ice have different impacts on the ocean, atmosphere, and ecosystem because of their different dynamic and thermodynamic properties. To date, fast ice and pack ice were researched as one sea ice entity (Cavalieri and Parkinson, 2008; Comiso et al., 2017; Parkinson and Cavalieri, 2012). The data in Table 4 show that the landfast ice area accounts for approximately 3%–4% of the total sea ice area in November during the study years, which is lower than the 5% estimated more than two decades ago (Fedotov et al., 1998). The ratios of the landfast ice area to the total sea ice area in 2007 and 2016 are higher than in other years (Table 4). In 2007, the high ratio of 3.92% was caused by the large landfast ice area whereas the highest ratio of 4.12% in 2016 resulted from the abnormally low pack ice area (Turner et al., 2017). The PO sector has the highest ratio of

Table 4
Landfast ice area and its ratio to total sea ice area in the five sectors of the Southern Ocean as well as the West and East Antarctic regions.

Year	Area of landfast ice in each sector ($\times 10^4$ km^2)									Ratio between landfast ice and sea ice areas (%)					
	WS	BAS	IO	PO	RS	Total	EA	WA	WA/EA	WS	BAS	IO	PO	RS	Total
2006	4.92	8.07	17.09	11.68	8.56	50.32	33.88	16.39	0.48	0.99	5.61	5.69	13.69	2.38	3.62
2007	6.20	7.08	17.12	16.36	8.94	55.70	38.86	17.37	0.45	1.18	4.45	6.88	21.80	2.20	3.92
2008	5.38	5.77	18.22	13.58	8.44	51.39	36.78	14.60	0.38	0.94	2.63	5.34	14.05	2.03	3.64
2009	6.13	6.00	16.86	15.30	6.00	50.29	36.59	14.03	0.46	1.32	3.04	5.55	19.89	1.77	3.64
2010	4.37	5.86	15.8	10.17	7.81	44.01	30.24	13.76	0.40	0.99	2.83	4.73	16.12	2.12	3.10
2011	3.94	7.38	17.05	12.56	8.87	49.80	35.47	14.33	0.38	0.73	4.85	6.75	12.77	2.27	3.48
2016	3.17	7.72	14.91	14.54	7.40	47.74	34.62	13.30	0.48	0.72	5.40	7.56	22.85	2.37	4.12
2017	4.01	8.73	14.88	11.64	7.40	46.66	31.42	15.23	0.43	0.96	4.40	6.22	12.53	2.24	3.65
Mean	4.77 ± 1.02	7.08 ± 1.04	16.49 ± 1.1	13.23 ± 1.96	7.93 ± 0.93	49.49 ± 3.25	34.73 ± 2.67	14.88 ± 1.30	0.43 ± 0.04	0.98	4.15	6.09	16.71	2.17	3.64

WS: Weddell Sea, BSA: Bellingshausen and Amundsen Seas, IO: Indian Ocean, PO: Pacific Ocean, RS: Ross Sea, EA: East Antarctica (10°W–170°E), WA: West Antarctica (170°E–10°W). The WA/EA is the ratio of the areas of WA and EA.

16.71% (averaged over all of the studied years) with a maximum of 22.85% in 2016. This is remarkably higher than the ratio for the entire Antarctic (3.64%). It should be noted that the PO sector has the second highest landfast ice area but has the smallest sea ice area among those five sectors On the other hand, the minimum ratio (average for all years) is found in the WS sector (0.98%) because of its expansive sea ice area, yet with the smallest landfast ice area among these five sectors.

To explore a possible relationship between the areas of landfast ice and total sea ice, the correlation coefficients between the areas were calculated during the study period. The landfast ice area in the RS and IO sectors has a high correlation coefficient (r) with the total sea ice area, 0.72 and 0.60 respectively. The correlation is lower in the WS sector with $r = 0.43$, while no correlation is found in the PO sector ($r = -0.13$). In the BA Seas, the correlation is negative ($r = -0.58$), which indicates that the landfast ice increases while sea ice decreasing. However, an analysis of the statistical significance shows that only the correlation of the landfast ice and sea ice areas in the Ross Sea sector is significant statistically, with a p-value of 0.0437 (<0.05). For other four sectors, their p-values are >0.05, which show that there is no significant correlation between fast ice and sea ice areas in these four sectors. In general, the correlation between landfast ice and sea ice areas in each sector is not strong, and the relationship has obvious regional difference.

4.3. Comparison against results from previous studies

Table 5 shows a comparison of landfast ice area from our method against areas reported in previous studies. Landfast ice in East Antarctica was detected using Radarsat-1 data acquired in November 1997 and 1999 for latitude ranges 87.8°E–170.3°E and 75.15°E–150.61°E (Giles et al., 2008). For the 87.8°E–170.3°E sector, the landfast ice area from the 1997 Radarsat-1 data is the lowest (162,788 km^2). It is similar to the area from our method only in 2010. For the rest of the years, the area varies between 1.63×10^5 km^2 to 2.23×10^5 km^2 with no apparent trend. For the 75.15°E–150.61°E sector, the 1999 data from Radarsat-1 shows an area of 1.6807×10^5 km^2, which matches with the area from our results in most years except 2007–2009 when our results show a higher area (ranging from 2.10×10^5 km^2 to 2.25×10^5 km^2).

Landfast ice was also detected using MODIS data from 2000 to 2008 for the East Antarctica (10°W–172°E) by Fraser et al. (2012). The author's results of the three overlapping (2006–2008) years are compared against our results in Table 4. Fraser's results during DOY 301–320 are used for 2006 and 2008 and those during DOY 281–300 are used for 2007. This was to match the acquisition time of SAR data in our study

Table 5
Estimates of landfast ice area (km^2) in different years (km^2)[a].

Year	Our results		Our results	Fraser's results
	Longitude over 87.8°E–170.3°E	Longitude over 75.15°E–150.61°E	Longitude over 10°W–172°E	
2006	185,699	175,882	339,047	336,248
2007	235,041	224,167	385,570	384,864
2008	199,212	210,475	367,753	363,920
2009	217,424	216,937		
2010	163,158	148,465		
2011	202,423	188,533		
2016	203,110	187,119		
2017	180,201	160,435		
Radarsat-1 (1997)	162,788	–		
Radarsat-1 (1999)	–	168,068		

[a] The landfast ice area from Radarsat-1 data of 1997 and 1999 in the table was acquired from published papers (Giles et al., 2008).

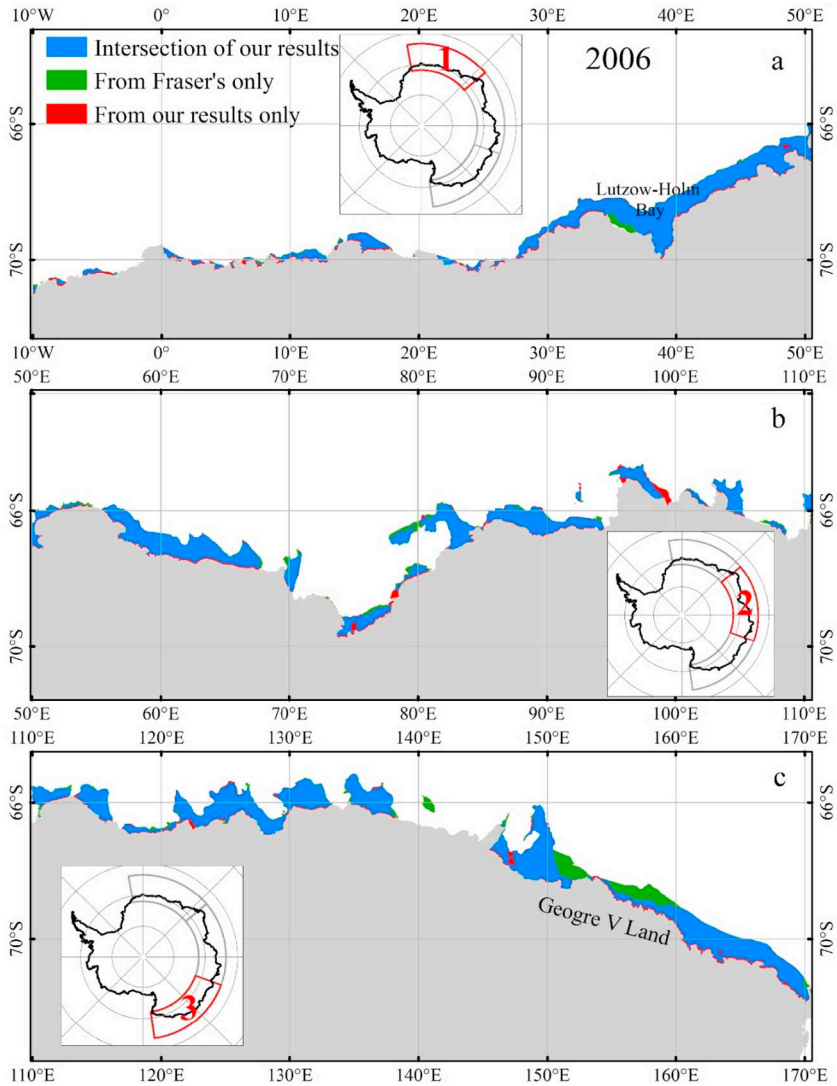

Fig. 6. Mapping of landfast ice off East Antarctic (10°W–172°E) in 2006 for our result and for Fraser's results in three sections. Red frames in insets indicate location of each section along the Antarctic coast. Blue areas represent the landfast ice which was detected by both our method and Fraser's method. Green areas represent the landfast ice detected only by Fraser's method and the red areas are landfast ice detected only by our method. Land is shown in grey color. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

as much as possible. The percentage difference of fast ice areas between the two data sets is about 0.83%, 0.18%, and 1.05% for 2006, 2007 and 2008, respectively. Although these difference are small difference, the landfast area from our results is higher for these three years. The difference may be attributed to the finer resolution of ASAR (150 m), which can detect more small-scale fast ice than the 1-km resolution of MODIS imagery used in Fraser et al. (2012). However, some significant differences are observed in certain sections as shown in Fig. 6. The figure shows the mapping of fast ice from our method and Fraser's method in 2006. In general, the fast ice in the east Antarctic resulting from SAR-derived overlaps well with the MODIS-derived results, although large differences are encountered in some areas.

The eastern Antarctic is divided into three sections to highlight significant differences. They are marked 1, 2 and 3 corresponding to panels (a), (b) and (c) in Fig. 6. In Section 1, there is a significant difference is represented by the green area near the shoreline around the Lutzow-Holm Bay. This is caused by the different coastlines used in our study versus Fraser's as shown in Fig. 7. In Fraser's method (Fraser et al., 2012), the MOA coastline (2004) was used (the red line in Fig. 7) while in our data the coastline from (Yu et al., 2019) was used (the green line in Fig. 7). This situation also exists at the same location in 2007 and 2008 because Fraser et al. (2012) used the same 2004 Antarctic coastline for 2002–2008. An annually updated coastline should be used to produce more accurate area of fast ice. Section 2 is an example of no significant change between the landfast ice areas from our method and Fraser's method. In Section 3, there are two obvious areas of remarkable difference along the George V Land (Fig. 6c, 150°–160°E). Unlike the case explained in Fig. 6a, the difference here is related to the outer border of the fast ice. This is caused by the differences in data and methods used in our and Fraser's studies. In Fraser's study, the main dataset used was a series of 20-day cloud-free composite MODIS images. The composite images represent a 20-day cloud-free mean image of the surface. The ice which is attached to the coast could be manually classified as fast ice; however, a problem may arise when the pack ice is advected against the fast ice, leaving no fracture or shear zone between itself and fast ice. It becomes difficult to distinguish the fast ice from pack ice in such as case (Fraser et al., 2010). In our method, on the other hand, three consecutive SAR images were used to calculate the spatial and temporal gradients at each pixel. With this method, regions with dynamic ice covers during the calculation period will be excluded from the fast ice identification regardless of the presence of leads or the shear zone attached to the fast ice. The

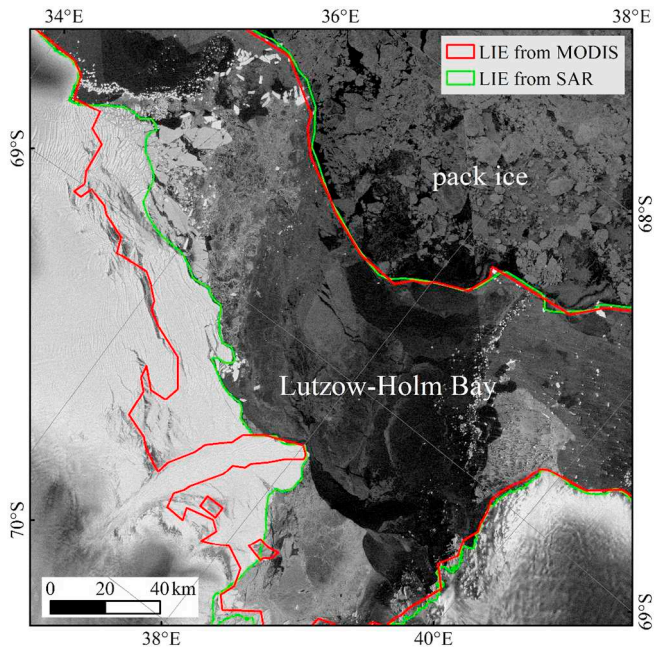

Fig. 7. Comparison between the LIEs derived from this study (green line) and the LIEs derived by Fraser et al. (2010) (red line) in Lutzow-Holm Bay. Note the significant difference in inner border of the fast ice (i.e., coastline) between the two methods. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

aforementioned difference, with the same explanation, is also found at 66.5°E, 105°E, 116°E and 150°E in 2007 and at 90°E and 98°E in 2008.

5. Case studies on interaction between landfast ice and icebergs

An iceberg may function as an anchor point for fast ice formation and retention. In SAR images, icebergs have a higher radar backscatter intensity than their surroundings (i.e., sea ice and water) because of microwave scattering by numerous air bubbles in the heterogeneous volume (Wesche and Dierking, 2012; Young et al., 1998). Based on iceberg identification in SAR imagery, we identified 24 zones where groups of icebergs were present within the landfast ice (dashed boxes in Fig. 4). In general, large icebergs can significantly affect the formation and extent of fast ice. The following two case studies address the interaction between fast ice and icebergs.

Case 1. Impact of iceberg on fast ice formation.

It is possible that sea ice becomes attached to an iceberg, forming a fast ice regime. In November 2008, an abnormal expansion of landfast ice was observed in the Barrier Bay in the eastern Prydz Bay, East Antarctica. The Barrier Bay is an open bay formed by the coast and the western end of the West Ice Shelf (Fig. 8). Many icebergs of various sizes are usually grounded in the north of the bay, and often cause the formation of a long west-east fast ice belt in winter. A polynya is usually formed in the bay in winter caused by the strong katabatic wind (Nihashi and Ohshima, 2015).

In November 2008, the Barrier Bay Polynya was covered by consolidated landfast ice and no open water existed. ASAR images (Fig. 8a) obtained in November 2008 show that a large iceberg B15B with a length of about 73 km and a smaller iceberg B15R were both grounded on the northwest section of the bay (77°31′45″E, 67°26′17″S) with their major axis orientation aligned north-south. The northern end of B15B had connected with the landfast ice belt in Four Ladies Bank where the bathymetric isobath is <300 m.

Based on the iceberg track data provided by Brigham Young University (Budge and Long, 2018), both the iceberg B15B and B15R arrived in Prydz Bay in October 2006 and continued to move until they became grounded in the west of Barrier Bay in October 2007. They drifted out of Prydz Bay in February 2009. Generally, sea ice formed in Barrier Bay Polynya is usually blown out quickly by the east or southeast wind and then new ice re-forms. During sea ice formation, brine in seawater is released into ocean, leading to formation of high salinity and dense water, which is important for bottom water formation. However, owing to the existence of the grounded icebergs B15B and B15R, sea ice formed in the Barrier Bay Polynya was blocked in and became difficult to drift out of the polynya, especially under weak wind conditions. Therefore, in November 2008, landfast ice, which had been formed earlier in the cold season, covered the entire bay (preventing the polynya formation). This only occurred in 2008 but not appear in 2007. The icebergs grounded there in October 2007 after the seasonal peak for the landfast ice formation. In 2009, the iceberg B15B and B15R drifted off to the north and the polynya reappeared in the Barrier Bay (Fig. 8b). However, a fast ice belt attached to the iceberg D15 remained in the Four Ladies Bank. This case study has shown the impact of the presence of offshore iceberg on the extent of fast ice.

Case 2. Offshore distance of icebergs can determine landfast ice distributions.

A special landfast ice zone attached to iceberg A23A was identified in front of the Filchner Ice Shelf (approximately 44°W) during 2006 to 2009 (Fig. 9). The giant iceberg A23A, which is located at the north of the landfast ice area, can be identified in ASAR images. The figure shows the LIE number of years and the progressive location of the iceberg (in inset). Iceberg A23A can fasten the sea ice as an anchor point. The offshore distance of the landfast ice increased from 180 to 190 km during 2006–2009. During these years, the northern boundary of the LIE was attached to the southern edge of A23A (Fig. 9). Starting from 2010, as the iceberg moved a small distance further to the north, however, the LIE offshore distance decreased sharply to 110 km. This distance decreased even further to approximately 35 km and 40 km in 2011 and 2016, respectively. In 2017, it increased again, expanding 120 km to north.

The results showed that the distance between the southern edge of iceberg A23A and the coastline was 180–190 km between 2006 and 2009. As the iceberg continued to move north, this distance increased to 194 km between 2009 and 2010, 210 km between 2010 and 2016, and 218 km between 2016 and 2017. The northern contour of the LIE remained attached to the southern contour of the iceberg; however, this changed in 2010, when the extent of the LIE became limited to 110 km away from the coastline. This indicated that the bonding of fast ice to the iceberg decreased or disappeared when the distance between the iceberg and coast exceeded a certain threshold. The landfast ice cannot advance beyond the threshold as numerous forces (e.g. coastal current, winds, ocean waves) contribute to its breakup. In this case, a distance 194 km may be the key threshold that maintains the landfast ice. This case demonstrates that icebergs can play a significant role in the development of the landfast ice in the Antarctic.

6. Conclusions

In this study, the LIE around Antarctica was identified based on the improved net gradient difference algorithm. We used 2470 scenes of C-band SAR data from ENVISAT and Sentinel-1 A/B to delineate the LIE around Antarctica during 2006–2011 and 2016–2017. MODIS data was used for validation and filling gaps in SAR coverage. Coastline contour from Yu et al. (2019) and MOA were used as the inner boundary of the landfast ice.

Validation was conducted in four regions. The results showed that the LIE matched well with the manually derived edge from MODIS visible images. Comparisons with previous data showed more landfast ice during the study period than in 1997 and 1999 for the region off East Antarctica. While the landfast ice area from this study is generally

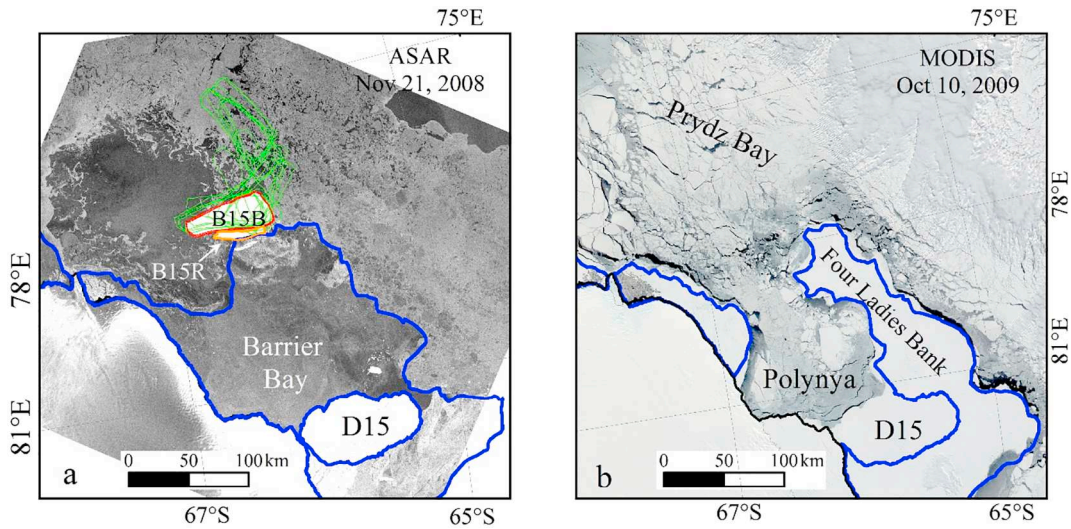

Fig. 8. (a) The LIE in 2008. Background ASAR image was acquired on November 21, 2008. Green outlines are of iceberg B15B derived from ASAR and MODIS images from January 2007 to September 2007. Red outlines are of B15B derived from MODIS image from October 2007 to November 2008 (remained stationary). Orange outlines are of B15R derived from MODIS images from October 2007 to November 2008. (b) LIE in 2009 overlaid on MODIS image acquired on October 10, 2009. Polynya reappeared after B15R and B15B drifted away. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

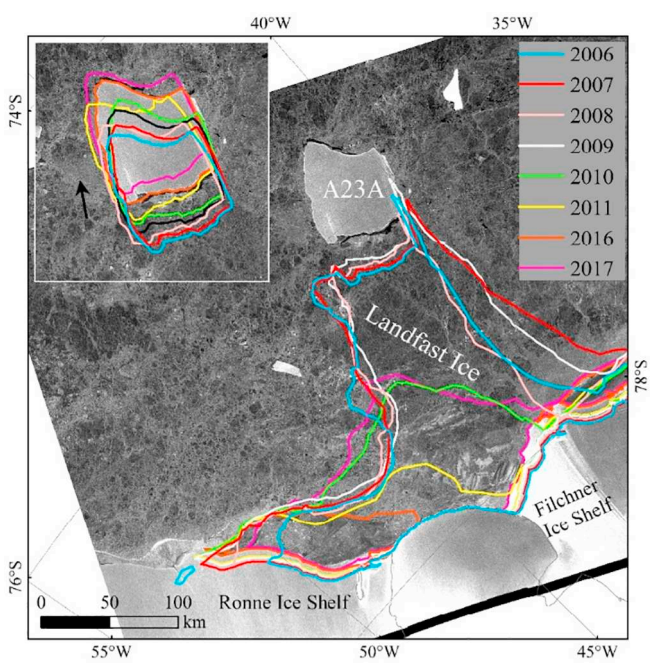

Fig. 9. LIE during 2006–2011 and 2016–2017. Background image is S1A image obtained on November 23, 2016. Inset shows the outlines of iceberg A23A derived from ASAR and Sentinel-1A/1B data in different years marked colors as shown in the color panel. Black arrow indicates the approximate motion direction of the iceberg A23A.

consistent with findings in Fraser et al. (2012) (based on MODIS composite), 2% more area is found in the East Antarctica.

Landfast ice area in November covers approximately $49.49 \pm 3.25 \times 10^4$ km^2 (average from eight years) accounting for approximately 3%–4% of the total Antarctic sea ice area. The landfast ice area in West Antarctica is approximately 38%–48% of that in East Antarctica. That is probably due to the existence of many narrow bays, grounded icebergs in east. During the study period, most of the recurrent landfast ice appeared off East Antarctica. For entire Antarctica, the width of landfast ice ranged from a few kilometers to >200 km.

Regionally, the landfast ice area is greatest in the Indian Ocean sector with a mean value of $16.49 \pm 1.1 \times 10^4$ km^2, while the minimum landfast ice area occurs in the Weddell Sea where only $4.77 \pm 1.02 \times 10^4$ km^2 exists on average. The average landfast ice extent in the Pacific Ocean sector is $13.23 \pm 1.96 \times 10^4$ km^2. This is the second largest sector among the five examined sectors, however, the ratio of landfast ice to sea ice area is the highest here because of less sea ice in this sector. Within the eight years the maximum ratio reached >20% in 2007 and 2016 in this sector. In the Weddell Sea sector, the average ratio is <1%. In general, the distributions of landfast ice in the Antarctic have significant regional difference.

Twenty-four landfast ice zones associated with groups of small grounded icebergs were identified. Most of these iceberg-landfast ice zones were located off East Antarctica, especially along the coasts of Wilkes Land and Oates Land. This implies that small, grounded icebergs would play a significant role in the formation of landfast ice and its maintenance in the Antarctic. Two cases are presented to illustrate how giant, grounded icebergs may affect landfast ice. The first case shows the impact of the iceberg on the formation of fast ice, and the second shows how the fast ice area and distribution change when the iceberg resumes its drift. To explore this relationship between icebergs and landfast ice, further more detailed studies could be conducted.

The present study has produced the first high-resolution and long time series of the LIE dataset for the Antarctic, which we used to carry out a spatial-temporal patterns analysis. We anticipate that this information can be used in support of studies of the Antarctic climate and ecology system.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Fengming Hui: Conceptualization, Supervision, Reviewing and Editing

Xiao Cheng: Conceptualization, Supervision, Reviewing and Editing

Zhaohui Chi: Data analysis, Reviewing and Editing

Petra Heil: Data analysis, Reviewing and Editing, Language polishing

Zhuoqi Chen: Data analysis, Reviewing and Editing

Yining Yu: Data analysis

Mengxi Zhai: Reviewing and Editing

References

Aoki, S., 2017. Breakup of land-fast sea ice in Lützow-Holm Bay, East Antarctica, and its teleconnection to tropical Pacific sea surface temperatures. Geophys. Res. Lett. 44, 3219–3227.

Beaman, R.J., Harris, P.T., 2005. Bioregionalization of the George V Shelf, East Antarctica. Cont. Shelf Res. 25, 1657–1691.

Bin, Cheng, Vihma, Timo, Zhanhai, Zhang, Zhijun, Li, Huiding, W., 2008. Snow and sea ice thermodynamics in the Arctic: model validation and sensitivity study against SHEBA data. Chinese Jounal of Polar Science 19, 108–122.

Budge, J.S., Long, D.G., 2018. A comprehensive database for Antarctic iceberg tracking using scatterometer data. IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing 11, 434–442.

Cavalieri, D.J., Parkinson, C.L., 2008. Antarctic Sea ice variability and trends, 1979–2006. Journal of Geophysical Research: Oceans 113 (C07004).

Comiso, J.C., Gersten, R.A., Stock, L.V., Turner, J., Perez, G.J., Cho, K., 2017. Positive trend in the Antarctic Sea ice cover and associated changes in surface temperature. J. Clim. 30, 2251–2267.

Enomoto, H., Nishio, F., Warashina, H., Ushio, S., 2002. Satellite observation of melting and break-up of fast ice in Lutzow-Holm Bay, East Antarctica. Polar Meteorology & Glaciology 16, 1–14.

ESA, 2007. ASAR Product Handbook.

ESA, 2016. Sentinel-1 Product Definition.

Fedotov, V.I., Cherepanov, N.V., Tyshko, K.P., 1998. Some features of the growth, structure and metamorphism of East Antarctic landfast sea ice. In: Jeffries, M.O. (Ed.), Antarctic Sea Ice: Physical Processes, Interactions and Variability. Antarctic Sea Ice: Physical Processes, Interactions and Variability AGU, Washington, D. C, pp. 343–354.

Ferrigno, J.G., Williams Jr., R.S., Foley, K.M., 2005. Coastal-change and glaciological map of the Saunders Coast area, Antarctica; 1972–1997. Ann. Glaciol. 39, 245–250.

Ferrigno, J.G., Cook, A.J., Mathie, A.M., Williams Jr., R.S., Charles, S., Foley, K.M., Fox, A.J., Thomson, J.W., Sievers, J., 2008. Coastal-change and glaciological map of the Larsen Ice Shelf area, Antarctica: 1940–2005. In: U.S. Geological Survey Geologic Investigations Series Map I–2600–B, 1 Map Sheet, (28-p. text).

Fraser, A.D., Massom, R.A., Michael, K.J., 2010. Generation of high-resolution East Antarctic landfast sea-ice maps from cloud-free MODIS satellite composite imagery. Remote Sens. Environ. 114, 2888–2896.

Fraser, A.D., Massom, R.A., Michael, K.J., Galton, B.K., Fenzi, Lieser, J.L., 2012. East Antarctic landfast sea ice distribution and variability, 2000–08. J. Clim. 25, 1137–1156.

Fretwell, P., Pretchard, H.D., Vaughan, D.G., et al., 2013. Bedmap2: improved ice bed, surface and thickness datasets for Antarctica. Cryosphere 7, 375–393.

Giles, A.B., Massom, R.A., Lytle, V.I., 2008. Fast-ice distribution in East Antarctica during 1997 and 1999 determined using RADARSAT data. Journal of Geophysical Research Oceans (C02S14), 113.

Heil, P., 1996. Seasonal and interannual variations of the oceanic heat flux under a landfast Antarctic Sea ice cover. J. Geophys. Res. 101, 25741–25752.

Heil, P., 2006. Atmospheric conditions and fast ice at Davis, East Antarctica: a case study. J. Geophys. Res. 111 (C05009).

Hunke, E.C., Lipscomb, W.H., Turner, A.K., 2010. Sea-ice models for climate study: retrospective and new directions. J. Glaciol. 56, 1162–1172.

Kim, S., Saenz, B., Scanniello, J., Daly, K., Ainley, D., 2018. Local climatology of fast ice in McMurdo Sound, Antarctica. Antarct. Sci. 30, 1–18.

Langhorne, P.J., Hughes, K.G., Gough, A.J., Smith, I.J., Williams, M.J.M., Robinson, N.J., Stevens, C.L., Rack, W., Price, D., Leonard, G.H., Mahoney, A.R., Haas, C., Hskell, T.G., 2015. Observed platelet ice distributions in Antarctic Sea ice: an index for ocean-ice shelf heat flux. Geophys. Res. Lett. 42 (13), 5442–5451.

Lei, R., Li, Z., Cheng, B., Zhang, Z., Heil, P., 2010. Annual cycle of landfast sea ice in Prydz Bay, east Antarctica. J. Geophys. Res. 115 (C02006).

Li, T., Liu, Y., Cheng, X., Ouyang, L.X., Li, X.Q., Liu, J.P., Shokr, M., Hui, F.M., Zhang, J., Wen, J.H., 2017. The effect of seafloor topography in the Southern Ocean on tabular iceberg drifting and grounding. Science China Earth Sciences 60, 697–706.

Li, X., Ouyang, L., Hui, F., Cheng, X., Shokr, M., Heil, P., 2018. An improved automated method to detect landfast ice edge off Zhongshan Station using SAR imagery. IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing 11, 4737–4746.

Linial, G., 2015. An apparent population decrease, or change in distribution, of Weddell seals along the Victoria Land coast. Marine Mammal Science 31, 1338–1361.

Mahoney, A., Eicken, H., Graves, A., Shapiro, L., Cotter, P., 2004. Landfast sea ice extent and variability in the Alaskan Arctic derived from SAR imagery. In: Geoscience and Remote Sensing Symposium, 2004. IGARSS '04. Proceedings. 2004 IEEE International. vol.2143. pp. 2146–2149.

Mahoney, A., Gaylord, A.G., Shapiro, L., 2007. Alaska landfast sea ice: links with bathymetry and atmospheric circulation. J. Geophys. Res. 112 (C02001).

Marshall, J., Speer, K., 2012. Closure of the meridional overturning circulation through Southern Ocean upwelling. Nat. Geosci. 5, 171–180.

Massom, R.A., Hill, K.L., Lytle, V.I., Worby, A.P., Paget, M.J., Allison, I., 2001. Effects of regional fast-ice and iceberg distributions on the behaviour of the Mertz Glacier polynya, East Antarctica. Ann. Glaciol. 33, 391–398.

Massom, R.A., Hill, Katrina, Barbraud, Christophe, Adams, Neil, Ancel, André, Emmerson, Louise, Pook, M.J., 2009. Fast ice distribution in Adélie Land, East Antarctica: interannual variability and implications for emperor penguins *Aptenodytes forsteri*. Mar. Ecol. Prog. Ser. 374, 243–257.

Massom, R.A., Giles, A.B., Fricker, H.A., Legresy, B., Warner, R.C., Hyland, G., Young, N., Fraser, A.D., 2010. Examining the interaction between multi-year landfast sea ice and the Mertz Glacier Tongue, East Antarctica: another factor in ice sheet stability? Journal of Geophysical Research Oceans 115 (C12027).

Miles, B.W.J., Stokes, C.R., Jamieson, S.S.R., 2017. Simultaneous disintegration of outlet glaciers in Porpoise Bay (Wilkes Land), East Antarctica, driven by sea ice break-up. Cryosphere 11, 1–37.

Murphy, E.J., Clarke, A., Symon, C., Priddle, J., 1995. Temporal variation in Antarctic sea-ice: analysis of a long term fast-ice record from the South Orkney Islands. Deep Sea Research Part I Oceanographic Research Papers 42, 1045–1062.

Nihashi, S., Ohshima, K.I., 2015. Circumpolar mapping of Antarctic coastal polynyas and landfast sea ice: relationship and variability. J. Clim. 28, 3650–3670.

Parkinson, C.L., Cavalieri, D.J., 2012. Antarctic sea ice variability and trends, 1979–2010. Cryosphere 6, 871–880.

Price, D., Rack, W., Langhorne, P.J., Haas, C., Leonard, G., Barnsdale, K., 2014. The sub-ice platelet layer and its influence on freeboard to thickness conversion of Antarctic Sea ice. Cryosphere 8, 1031–1039.

Scambos, T.A., Haran, T.M., Fahnestock, M.A., Painter, T.H., Bohlander, J., 2007. MODIS-based Mosaic of Antarctica (MOA) data sets: continent-wide surface morphology and snow grain size. Remote Sens. Environ. 111, 242–257.

Schubert, A., Small, D., Miranda, N., Meier, E., 2008. ASAR product consistency and geolocation accuracy. In: CEOS SAR Calibration and Validation Workshop 2008, Oberpfaffenhofen, 27 November 2008–28 November 2008, (online).

Schubert, A., Miranda, N., Geudtner, D., Small, D., 2017. Sentinel-1A/B combined product geolocation accuracy. Remote Sens. 9 (6), 607.

Spreen, G., Kaleschke, L., Heygster, G., 2008. Sea ice remote sensing using AMSR-E 89-GHz channels. J. Geophys. Res. 113 (C02S03).

Tamura, Takeshi, Ohshima, Kay I., Nihashi, S., 2008. Mapping of sea ice production for Antarctic coastal polynyas. Geophys. Res. Lett. (L07606), 35.

Trathan, P.N., Fretwell, P.T., Bernard, S., 2011. First recorded loss of an emperor penguin colony in the recent period of Antarctic regional warming: implications for other colonies. PLoS One 6 (2), e14738.

Turner, J., Phillips, T., Marshall, G.J., Hosking, J.S., Pope, J.O., Bracegirdle, T.J., Deb, P., 2017. Unprecedented springtime retreat of Antarctic Sea ice in 2016. Geophys. Res. Lett. 44, 6868–6875.

Ushio, S., 2006. Factors affecting fast-ice break-up frequency in Lützow-Holm Bay, Antarctica. Ann. Glaciol. 44, 177–182.

Wesche, C., Dierking, W., 2012. Iceberg signatures and detection in SAR images in two test regions of the Weddell Sea, Antarctica. J. Glaciol. 58, 325–339.

Wongpan, P., Hughes, K.G., Langhorne, P.J., Smith, I.J., 2018. Brine convection, temperature fluctuations and permeability in winter Antarctic land-fast sea ice. Journal of Geophysical Research Oceans 123 (3), 216–230.

Young, N., Turner, D., Hyland, G., Williams, R., 1998. Near-coastal iceberg distributions in East Antarctica, 50–145° E. Ann. Glaciol. 27, 68–74.

Yu, Y., Zhang, Z., Shokr, M., Hui, F., Cheng, X., Chi, Z., Heil, P., Chen, Z., 2019. Automatically extracted Antarctic coastline using remotely-sensed data: an update. Remote Sens. 11 (16), 1844.

Zan, F., Guarnieri, A.M., 2006. TOPSAR: terrain observation by progressive scans. IEEE Trans. Geosci. Remote Sens. 44, 2352–2360.

Zhao, J., Yang, Q., Bin, C., Wang, N., Hui, F., Shen, H., Han, X., Zhang, L., Timo, V., 2017. Snow and land-Fast Sea ice thickness derived from thermistor chain buoy in the Prydz Bay, Antarctic. Haiyang Xuebao 39, 115–127.

Zhou, N., Zhengxu, Zhao, Suwen, W., 2010. Medium edge detection method in image. Journal of Image and Graphics 15, 397–402.